6COSC023W

Computer Science Final Project

Project Proposal and Requirements Specification (PPRS) Handout

AN ADAPTIVE, CONFIDENCE-WEIGHTED VERIFICATION FRAMEWORK FOR MITIGATING LARGE LANGUAGE MODEL HALLUCINATIONS

Student: Mr. Shaun Yogeshwaran (20210069)

Supervisor: Mr. John Sriskandarajah

Degree: BSc/BEng (Hons) Computer Science

School of Computer Science & Engineering

College of Design, Creative and Digital Industries

University of Westminster

13 November 2025

Declaration

This document has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

Student Name: Shaun Geethapriyan Yogeshwaran

Date of Submission: 13/11/2025

1. Definition of the Problem

- **Problem Description**: The most significant issue with large language models (LLMs) is known as "hallucination," which means they can generate text that sounds plausible and even fluent, but is entirely false. Since LLMs can produce entirely fabricated information, their lack of reliability is the largest barrier to deploying them in "high stakes" applications like healthcare, law, and finance. These are areas where producing accurate information is crucial, so a single "hallucinated" piece of information could lead to disastrous consequences (i.e. an inaccurate diagnosis or fabrication of a legal ruling).

  Although some solutions, like retrieval-augmented generation (RAG) and logical-based verifiers (i.e. chain of natural language inference-CONLI), improve the factuality of LLMs, they each have a similar, unresolved flaw: the uniform threshold constraint. Uniform thresholds mean that the single, one-size-fits-all threshold (for example: nli_entailment_score > .95) is applied to every generated statement, regardless of context or the quality of evidence used to support those statements.

  While this rigid application of uniform thresholds presents an inherent and suboptimal trade off:

  Over-flagging (High False Positives): Using a high threshold (for example, 0.99) to catch very slight or very fine-grained hallucinations would flag too many valid statements that are correct based on nuances, abstractions, and other forms of contextualization. This would result in very high false positive rates and a corresponding decrease in usability and user experience.

  Missed Hallucinations (High False Negatives): If you use a lower threshold (for example, 0.85) to allow for greater nuance and to minimize the likelihood of over-flagging, you would likely miss many true and subtle hallucinations.

  This is a problem that occurs because the verification layer has no knowledge about the quality of the evidence it uses. The upstream RAG component produces a very important "confidence score" (for example: cosine similarity) that quantitatively expresses the degree of trustworthiness of the retrieved evidence. The confidence score produced by RAG is usually ignored; and the same pass/fail criterion is applied to the verifier regardless of the degree of relevance of the

evidence used to support a statement (for example: 99% or 65%). The root cause of the problem is the disconnect between the RAG system's confidence signal and the decision making process of the verifier.

- **Significance**: This project has dual significance as it addresses important shortcomings in both the problem space and the research space:
- Problem Space Contribution: The most significant contribution of this work is developing the Confidence-Weighted CONLI (Cw-CONLI) algorithm to address the "Uniform Threshold Limitation," the primary shortcoming in current approaches to NLI-based verification. Cw-CONLI is an adaptive, "plug-and-play" verification system, which adjusts the verifier's strictness level by the RAG component's confidence in its retrieval results. In doing so, Cw-CONLI eliminates the fixed binary trade-offs of current systems and aims to create a more accurate, efficient and user-friendly verification pipeline for use in real world applications, by reducing both hallucinations (False Negatives) and over flagging (False Positives).
- Research Space Contribution: The contributions of this work will be in providing the first systematic, three-condition empirical analysis of the trade off between accuracy and latency of using dynamic vs. uniform thresholds for verification in NLI-based verification. This work will provide the AI and RAG development communities with empirical data validating the confidence weighted approach, to help inform the design of more intelligent and efficient verification systems through experimentation and evaluation using the HaluEval benchmark, by comparison of the performance of three different verification pipelines (C1: RAG-only Baseline, C2: RAG + Standard CONLI, and C3: RAG + Cw-CONLI).

2. Aims, Objectives, and Scope

- **Aim**: The aim of this research is to design, implement, and rigorously evaluate a two-layer hallucination mitigation pipeline (AFLHR Lite), centered on a novel Confidence-Weighted CONLI algorithm, to empirically measure its improvements in detection accuracy and latency trade-offs compared to uniform-threshold approaches.
- **Objectives**:
    - Develop a confidence-weighting algorithm that dynamically adjusts NLI verification thresholds based on RAG retrieval confidence scores.

- Integrate features such as a RAG component (using FAISS and all-MiniLM-L6-v2) and a CONLI verifier (using roberta-large-mnli) to implement the AFLHR Lite two-layer pipeline.
- Ensure the application can be rigorously evaluated by designing and executing a 3-condition experiment (Baseline vs. Standard-CONLI vs. Weighted-CONLI) on the HaluEval benchmark.
- Provide a quantitative analysis of the trade-offs between hallucination detection (F1-score), latency, and over-flagging (false positives).

- **Scope**:
  - **Inclusions:**

    The project will include functionalities such as:

    - The design, implementation, and evaluation of the AFLHR Lite pipeline, orchestrated using the LangChain library framework.
    - A RAG pipeline built from a curated corpus from the English Wikipedia, the sentence-transformers/all-MiniLM-L6-v2 embedding model, and an in-memory FAISS vector store.
    - A retrieval mechanism configured to extract both the source documents (premise) and their numerical confidence scores.
    - A core generative model (meta-llama/Meta-Llama-3-8B-Instruct) for all experimental conditions.
    - The programmatic Confidence-Weighted CONLI layer, using FacebookAI/roberta-large-mnli to generate scores and implementing the dynamic thresholding algorithm.
    - An evaluation harness to automatically run the 3-condition experiment on the HaluEval benchmark and compute all key metrics.

  - **Exclusions:**
    The project will not include:

    - A Cross-Model Consistency Layer (e.g., using a secondary, architecturally distinct model like Mistral-7B).
    - A Surgical Correction Layer to automatically correct detected hallucinations.
    - DPO Knowledge Distillation (a stretch goal deferred to future work).

3. Background Review

- **Key Findings (Literature / Systems)**:

A systematic review of the literature reveals a clear and un-bridged research gap. The field of hallucination mitigation has evolved through several key stages, each with distinct limitations:

1. Foundational RAG (Lewis et al., 2020): RAG provides a strong baseline for grounding LLMs in external, verifiable knowledge. However, RAG itself does not guarantee that the model will faithfully or logically use the retrieved context.
2. Flawed Self-Verification (Dhuliawala et al., 2024): Methods like Chain-of-Verification (CoVe) were proposed to have the LLM check its own work. These approaches fundamentally fail due to the "recursive self-correction fallacy," wherein an LLM cannot be an objective arbiter of its own (potentially flawed) reasoning.
3. Logic-Centric Verification (Lei et al., 2023): The emergence of NLI-based verifiers like CONLI provided a more objective, external check by using a specialized classifier to determine if a generated claim is logically entailed by the source text. However, all current implementations of this method rely on a single, rigid "uniform threshold," which leads directly to the over-flagging vs. missed-hallucinations trade-off.
4. The Un-Bridged Gap: A systematic literature search confirms that two fields of research-Hallucination Verification (e.g., CONLI) and Retrieval Calibration (e.g., SGIC by Chen et al., 2025)-remain separate. No published framework was found that uses RAG retrieval confidence scores to dynamically modulate the thresholds of a post-hoc NLI verifier. This project is the first to bridge this gap.

- Theoretical Foundation: The proposed solution is not an ad-hoc heuristic but is grounded in established machine learning theory, specifically Cost-Sensitive Learning. The Cw-CONLI algorithm can be understood as an inference-time implementation of this theory. It uses the retrieval score as a dynamic proxy for the misclassification cost:

- Low retrieval score: Implies low trust in the evidence. The cost of a False Negative (missing a hallucination) is high. The algorithm, therefore, sets a strict threshold (e.g., 0.99) to avoid this high-cost error.
    - High retrieval score: Implies high trust in the evidence. The cost of a False Positive (over-flagging a valid, nuanced claim) is now high, as it degrades user experience. The algorithm, therefore, sets a lenient threshold (e.g., 0.85) to avoid this error.

- **Comparison with Similar Software Applications/Products**
- The following table, summarizes the foundational mitigation strategies and their limitations, which this project directly addresses:

Citation	Summary	Limitation	Contribution
Lewis et al. (2020)	Introduced **Retrieval-Augmented Generation (RAG)** to ground LLM outputs in external knowledge, improving factuality.	Does not inherently verify if the model faithfully or logically uses the retrieved context.	Established the paradigm of combining parametric and non-parametric knowledge for generation.
Dhuliawala et al. (2024)	Proposed **Chain-of-Verification (CoVe)**, a self-verification method where an LLM generates a plan to check its own facts.	Fails to detect logical errors between facts; suffers from the "recursive self-correction fallacy".	Demonstrated the potential of self-correction but highlighted its core vulnerabilities.
Lei et al. (2023)	Introduced **Chain of Natural Language Inference (CONLI)** as a logic-centric method for hierarchical hallucination detection.	Introduces latency; standard implementations rely on rigid, **uniform verification thresholds**.	Provided a method for detecting hallucinations without external knowledge sources.
Manakul et al. (2023)	Introduced **SelfCheckGPT**, which uses sampling from the LLM to check for consistency and identify hallucinations.	Primarily detects inconsistencies within the model's own knowledge, not necessarily against ground truth or logic.	Provided a method for detecting hallucinations without external knowledge sources.

Differentiation:

The differentiation of this project (AFLHR Lite) is its singular focus on solving the uniform threshold limitation, which is the key, unaddressed limitation of state-of-the-art verifiers like CONLI.

While advanced RAG systems like SGIC (Chen et al., 2025) also use confidence scores, they use them to calibrate the retrieval and generation

process (i.e., to "write a better response"). They do not, however, address the post-hoc verification step.

This project is the first to "bridge the gap" between these two fields by:

1. Using the RAG retrieval confidence score (a signal all other verifiers waste).
2. To dynamically modulate the entailment threshold of the CONLI verifier (a new algorithmic mechanism).

This Confidence-Weighted CONLI algorithm directly addresses the "over-flagging" problem, creating a more efficient, practical, and intelligent verifier that adapts its strictness based on the quality of the evidence.

4. Tools (Hardware/Software)

The following hardware and software, as detailed in Table 1.3 of the full proposal, are required for the project:

Hardware :

Component	Specification
Processor (CPU)	Apple M2, 8-core
Memory (RAM)	8 GB Unified Memory
Storage	256 GB SSD
GPU / Compute	Cloud-based GPU (Google Colab / AWS EC2 / OpenAI API) VRAM of 20GB+

Software :

Component	Specification
Python and libraries	Version 3.10 or higher
LangChain	Latest stable release

Google Colab	Paid version
MS Word / Google Docs	Paid version
FAISS	Latest stable
VS Code	Latest version
Git & GitHub	Latest version

1. Use Cases and Diagrams

- **Use Cases**:
  - **Use Case:** Submit Prompt and Receive Verified Response
  - **Actors:** End-User, Developer / Integrator, AI Researcher
  - Description: An actor submits a query (prompt) to the system. The system executes its full two-layer pipeline, which includes the following key included use cases:
    - Retrieve Document Chunks: The RAG layer retrieves relevant documents (premise) and, crucially, their numerical confidence scores (retrieval_score).
    - Generate LLM Response: The Llama-3-8B model generates a response (hypothesis) based on the retrieved premise.
    - Apply Hypothesis w/ NLI: The hypothesis is broken into individual claims, and each is compared against the premise using the RoBERTa-NLI classifier, which generates an entailment_score.
    - Apply Dynamic Threshold: The Cw-CONLI algorithm uses the retrieval_score from Step 1 to dynamically select a strict or lenient entailment_score threshold.

2. Requirements Elicitation

- **Approach**:
  - **[Method 1]**: **Literature Review:**
    A systematic and comprehensive review of academic and technical literature served as the foundational elicitation method. This approach was used to:
  - Identify the state-of-the-art in LLM hallucination (RAG, CONLI).
  - Discover the "uniform threshold limitation" as the primary unresolved problem in existing verifiers
  - Confirm the "un-bridged research gap" between retrieval calibration and verification
  - Establish the theoretical grounding for the solution in cost-sensitive learning

  - **[Method 2]**: Prototyping: An iterative prototyping approach was used to validate technical assumptions and uncover implicit constraints. This method was crucial for discovering technical-level requirements.
  - Key Prototype Finding: Prototyping revealed that retrieval scores from the embedding model (all-MiniLM-L6-v2) are non-uniform and tend to cluster in narrow ranges (e.g., 0.75-0.95). This technical finding is a critical requirement, as it dictates that a simple linear mapping of scores to thresholds would fail. It directly justifies the design of a more robust tiered-threshold or calibrated weighting function for the Cw-CONLI algorithm.

Requirement Specification:
Functional Requirements (FR)

ID	Priority	Level	Requirement Description
FR01	MUST		The system must ingest a knowledge base and user query.
FR02	MUST		The RAG system must retrieve semantically relevant document chunks.

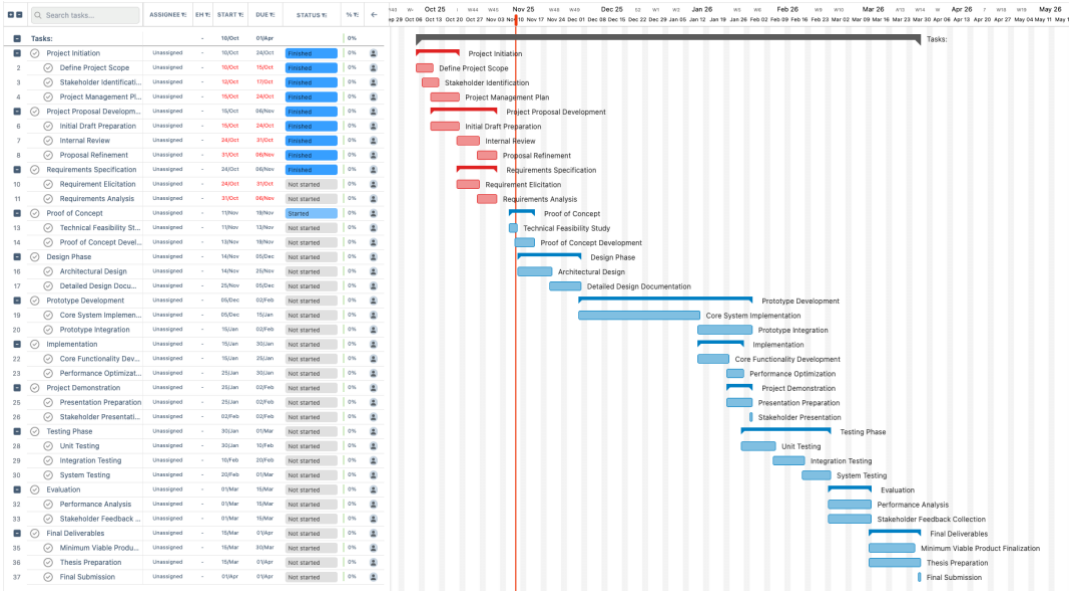
FR02.1	MUST		The RAG retriever must output both the document chunk (premise) and its numerical **$retrieval_score$**.
FR03	MUST		The system must use the Primary LLM (Llama-3-8B) to generate an initial response (hypothesis) based on the retrieved premise.
FR04	MUST		The system must verify the hypothesis against the premise using a pretrained NLI classifier (roberta-large-mnli).
FR05	MUST		The system must implement the **Confidence-Weighted CONLI algorithm**.
FR05.1	MUST		The system must apply a **dynamic NLI entailment threshold** based on the $retrieval_score$ from FR02.1.
FR06	MUST		The system must flag any sentence that fails the dynamic verification check as a REJECTED_AS_HALLUCINATION.
FR07	SHOULD		The system should provide a programmatic interface to run the evaluation harness (C1, C2, C3) on the HaluEval dataset (Li et al., 2023).
FR08	COULD		The system could implement an alternative continuous weighting function to compare against the tiered-threshold model.
FR09	WILL NOT HAVE		The system will not implement a cross-model consistency check (Deferred to future work).
FR10	WILL NOT HAVE		The system will not implement a **surgical correction layer** to rewrite flawed sentences (Deferred to future work).
FR11	WILL NOT HAVE		The system will not implement a DPO pipeline (Deferred to future work).

Non-Functional Requirements (NFR)

ID	Priority	Requirement and Description
NFR01	MUST	**(Reliability)** The system must produce deterministic verification outputs for the same (premise, hypothesis, score) triplet.
NFR02	MUST	**(Modularity)** The verification layer (Layer 2) must be modular, allowing the Standard-CONLI (C2) and Confidence-Weighted-CONLI (C3) components to be swapped for evaluation.
NFR03	MUST	**(Performance)** The system's performance (F1, Latency, Over-flagging) must be quantifiable on the HaluEval benchmark (Li et al., 2023).
NFR04	SHOULD	**(Usability)** The evaluation harness must be executable via a single script.
NDFr05	MUST	**(Resource Efficiency)** The full end-to-end inference pipeline (AFLHR Lite) must be able to run on a single GPU with < 20GB VRAM.

3. Time Schedule

- **Gantt Chart Overview**:



  - **Phase 1: Research and Planning (Weeks 1-4 / Oct-Nov 2025)**
    - [24th Oct 2025] : Project Proposal and Requirement Specification – Initial draft
    - [13th Nov 2025] : Project Proposal and Requirement Specification – Final
  - **Phase 2: Prototyping and Design (Weeks 5-8 / Nov-Dec 2025)**
    - [19th Nov 2025] : Proof of Concept
  - **Phase 3: Implementation and Evaluation (Weeks 9-12 / Jan-Mar 2026)**
    - [30th Jan 2026] : Implementation (Completion)
    - [2nd Feb 2026] : Prototype (Final) & Interim Project Demo
    - [1st March 2026] : Testing & Evaluation (Completion)
  - **Phase 4: Finalization (Weeks 13-14 / Apr 2026)**
    - [1st Apr 2026] : Thesis Submission
    - [1st Apr 2026] : Minimum Viable Product

4. References / Bibliography

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5. Ethics Form

UNIVERSITY OF WESTMINSTER

OFFICE USE: ____ - ___-_____

University of Westminster
University Research Ethics Committee

Application for Research Ethics

PART A

Section 1 – PROJECT AND APPLICANT DETAILS

1.1 **Project Title:**
An Adaptive, Confidence-Weighted Verification Framework for Mitigating Large Language Model Hallucinations

1.2 **Applicant Details**

Name: Shaun Geethapriyan Yogeshwaran	**University Email Address:** w1912919@my.westminster.ac.uk
Contact Address: 222, Angulana Station Road, Moratuwa, Sri Lanka.	**Telephone Number:** +94 0777202816
Faculty: Science and Technology	

Please check the relevant box:

Undergraduate ☑ **Postgraduate** ☐ **MPhil/PhD Student** ☐ **Staff** ☐

I confirm I have read the *University's Code of Practice Governing the Ethical Conduct of Research*	**YES** ☑	**NO** ☐

1.3 Supervisor/Dean of Faculty/Faculty Research Director details

Please note that all applicants with a supervisor(s) must ensure that the supervisor signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

All **staff** must ensure that their Dean of Faculty, or Faculty Research Director (or nominee), as appropriate, signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

Name: John Sriskandarajah	**University Email Address:** john.s@iit.ac.lk
Faculty: Science and Technology	**Telephone Number:** N/A

Shaun Y

PART A (Continued)

Section 2 – Project Details

2.1 Please provide a description of the background with references to relevant literature (250 words maximum):
Large Language Models (LLMs) have demonstrated unprecedented capabilities in language understanding and generation; however, they suffer from a critical limitation: their propensity to "hallucinate," generating plausible-sounding but factually incorrect information. This phenomenon—encompassing both intrinsic hallucinations (contradictions with source material) and extrinsic hallucinations (ungrounded claims)—represents a fundamental barrier to deploying LLMs in high-stakes domains such as healthcare, law, and finance, where factual accuracy is paramount.

Retrieval-Augmented Generation (RAG) has emerged as a foundational strategy to mitigate hallucinations by grounding model outputs in external knowledge sources. However, RAG alone does not guarantee faithful use of retrieved evidence. More sophisticated logic-centric verification methods, such as Chain of Natural Language Inference (CONLI), use specialized Natural Language Inference (NLI) classifiers to verify whether generated claims are logically entailed by retrieved source material.

Despite these advances, a critical limitation persists: existing verification systems rely on rigid, uniform thresholds. A static entailment threshold (e.g., score > 0.95) creates a suboptimal trade-off—strict thresholds catch subtle hallucinations but over-flag valid, nuanced responses, while lenient thresholds miss actual hallucinations. The root cause is that verification pipelines remain "blind" to evidence quality; they apply identical criteria regardless of retrieval confidence, a valuable but typically discarded signal from the RAG component.
This research gap—the absence of an integrated framework that uses retrieval confidence to dynamically calibrate verification strictness—motivates the proposed solution: a Confidence-Weighted CONLI algorithm that adapts verification thresholds based on evidence trustworthiness, enabling more efficient and accurate hallucination detection.

2.2. Please provide a brief description and the aims of your study (250 words maximum):

Study Description:

This research proposes an adaptive, confidence-weighted verification framework for mitigating hallucinations in Large Language Models (LLMs). Large Language Models demonstrate remarkable capabilities in language understanding and generation; however, they suffer from a critical limitation—their propensity to "hallucinate," generating plausible but factually incorrect information. Current hallucination mitigation strategies, such as Retrieval-Augmented Generation (RAG) and logic-based verifiers like Chain of Natural Language Inference (CONLI), improve factuality but rely on inflexible, uniform verification thresholds. This one-size-fits-all approach creates a suboptimal trade-off: strict thresholds catch subtle hallucinations but incorrectly reject valid, nuanced responses (over-flagging), while lenient thresholds miss actual hallucinations. The core issue is that verification pipelines remain "blind" to evidence quality; they apply identical criteria regardless of retrieval confidence—a valuable signal typically discarded from the RAG component. The proposed solution is the AFLHR Lite pipeline, featuring a novel Confidence-Weighted CONLI algorithm that dynamically calibrates verification strictness based on retrieved evidence trustworthiness.

Research Aims and Objectives:

The primary aim is to design, implement, and rigorously evaluate a two-layer hallucination mitigation pipeline centered on the Confidence-Weighted CONLI algorithm to empirically measure improvements in detection accuracy and latency trade-offs compared to uniform-threshold approaches. The specific objectives include: (1) designing a confidence-weighting algorithm that dynamically adjusts Natural Language Inference verification thresholds based on RAG retrieval confidence scores; (2) implementing the

AFLHR Lite two-layer pipeline with a success criterion of latency increase not exceeding 20% versus the standard-CONLI baseline; (3) conducting a rigorous 3-condition empirical evaluation on the HaluEval benchmark with a success criterion of achieving 5% F1-score improvement; and (4) analyzing empirical trade-offs between hallucination detection accuracy, latency, and over-flagging, with a success criterion of reducing false positives by 10% compared to the standard approach.

2.3. Please outline the design and methodology of your study (include details of the selection and recruitment of participants (if any) and details of any invasive (e.g. blood samples, inhalation/ingestion of food and/or non-food products (in abnormally higher or lower levels than normal or a different form), or intrusive (e.g. questionnaires, focus groups, interviews, etc.) procedures [attach extra information as necessary] (400 words maximum in total):

Study Design:

This research employs a pragmatic, deductive, experimental methodology structured around a rigorous 3-condition controlled experiment. The study compares three computational conditions using the HaluEval benchmark—a large-scale, human-annotated hallucination evaluation dataset specifically designed for assessing LLM hallucination detection performance.

Experimental Conditions:

- Condition 1 (C1 - Baseline): RAG-only approach with no verification layer
- Condition 2 (C2 - Control): RAG combined with standard CONLI verification using a uniform, static NLI entailment threshold (e.g., 0.95)
- Condition 3 (C3 - Experimental): RAG combined with the proposed Confidence-Weighted CONLI algorithm, which dynamically adjusts NLI thresholds based on retrieval confidence scores

Methodological Approach:
The core methodology is computational rather than involving human participants. The AFLHR Lite pipeline operates at inference time without human recruitment or participation. The system comprises two layers: (1) a Retrieval-Augmented Generation (RAG) layer using Meta-Llama-3-8B-Instruct as the generative model, sentence-transformers (all-MiniLM-L6-v2) for embeddings, and a FAISS vector store indexed on English Wikipedia; and (2) a Confidence-Weighted CONLI verification layer using a pretrained NLI classifier (FacebookAI roberta-large-mnli).

Evaluation Protocol:

The HaluEval benchmark dataset will be split into a development set for threshold tuning and a blind test set for final evaluation. A grid search will optimize parameters for both C2 and C3 on the dev set. The locked-in models will be executed once on the test set to generate final results. Key metrics include hallucination detection F1-score (primary), query latency (milliseconds), and false positive rate (over-flagging).

No Human Participants or Invasive Procedures:

This study does not involve human subjects, recruitment, or any invasive or intrusive procedures. It is a computational evaluation of machine learning models on a publicly available benchmark dataset. No biological samples, ingestion of substances, questionnaires, interviews, or focus groups are employed.

2.4. Timescales

Start Date (DD/MM/YY): 10/10/2025

Estimated duration of work: Approximately 120 days

Section 3 - RISK OF HARM

NOTE 1: Where indicated below applicants should check if the research will require ethical approval from a National Research Ethics Committee via the Integrated Research Application System (IRAS) - nres.queries@nhs.net.- http://www.hra-decisiontools.org.uk/ethics/
NOTE 2: The University of Westminster holds a Human Tissue Authority Licence – This licence is specifically for tissue stored at 115 New Cavendish Street in accordance with the terms of the licence – Advice must be obtained from the University Human Tissue Designated Individual (N.Presneau@westminster.ac.uk)

	RISK OF HARM (to self, colleagues, participants, environment or animals)	Yes	No	N/A
1	Will any pain or more than mild discomfort result from the study?	☐	☐	☑
2	Could the study induce any psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life?	☐	☐	☑
3	Will the study involve prolonged or repetitive physical or psychological testing of human participants that may put someone at risk, e.g. use of treadmill?	☐	☐	☑
4	Will the study involve raising sensitive topics (e.g. sexual activity, drug use, revelation of medical history, bereavement, illegal activities, etc.)?	☐	☐	☑
5	Does your work involve any "relevant material" containing human cells (e.g. blood, urine, saliva, body tissues but NOT established cell-lines) from living or deceased persons (Such work must take account of the Human Tissue Act)? – See Note 1 and 2 above.	☐	☐	☑
6	Will DNA samples be taken from human participants (Such work must take account of the Human Tissue Act)? – See Note 1 and 2 above.	☐	☐	☑
7	Does your study raise any issues of personal safety for you or other researchers or participants involved in the project (Especially relevant if taking place outside working hours or off University premises)?	☐	☐	☑
8	Does your study involve deliberately misleading the participants (e.g. deception, covert observation)?	☐	☐	☑
9	Does your work involve administration of a food or non-food substance of a different type from or in abnormally higher or lower amounts than normal or one that is known to cause allergic reaction(s) or potential psychological stress?	☐	☐	☑
10	Does your study involve issues relating to personal and/or sensitive data?	☐	☐	☑
	PARTICIPANTS (and/or their records/associated data) **Does your work involve any of the following:**	**Yes**	**No**	**N/A**
11	Human participants in a health and/or social care setting (e.g. patients, those attending day centres, community care, rehabilitation centres, etc., including in the NHS, other public, private and/or voluntary sectors)? – See Note 1 above.	☐	☐	☑
12	Human participants who may be deemed vulnerable (e.g. children, people in poverty and/or with physiological or psychological impairments, persons attending rehabilitation centres, persons in easily identifiable positions that could be subject to victimisation, etc.)?	☐	☐	☑
13	Expectant or new mothers?	☐	☐	☑
14	Refugees/Asylum seekers?	☐	☐	☑
15	Minors (under the age of 18 years old)?	☐	☐	☑
16	Participants in custody (e.g. prisoners or arrestees)? – See Note 1 above.	☐	☐	☑
17	Participants with impaired mental capacity (e.g. severe mental illness, brain damage, sectioned under Mental Health Act, lowered or reduced sense of consciousness)? – See Note 1 above.	☐	☐	☑
18	Animals (or animal tissue).	☐	☐	☑
	INFORMATION TO PARTICIPANTS	**Yes**	**No**	**N/A**
19	Will you provide participants with a Participant Information Sheet prior to obtaining informed consent which can be taken away by the participant?	☑	☐	☐
20	Will you describe the procedures to participants in advance, so that they are informed about what to expect?	☑	☐	☐
21	Will you obtain informed consent for participation (normally written)? OR in the case of using personal data previously acquired was consent given for the reuse of the data for other research purposes?	☑	☐	☐
22	Will you tell participants that they may withdraw from the research at any time and for any reason without any impact on their care, service provision etc.?	☑	☐	☐
23	Will you give participants the option of omitting questions they do not want to answer?	☑	☐	☐
24	Will you tell participants that their data will be treated as confidential and that, if published, it will not be identifiable as theirs?	☑	☐	☐

25	Will you offer feedback to participants at the end of their participation, upon request (e.g. give them a brief explanation of the study and its outcomes)?	☑	☐	☐
26	Has external funding or collaboration been applied for/received, which requires institutional ethical consideration or approval?	☐	☑	☐

Useful links:

- http://www.screc.org.uk/ - Social Care Research Ethics Committee
- http://www.hra-decisiontools.org.uk/ethics/ - Human Research Authority decision tool to identify if research needs National Research Ethics Committee approval
- http://www.nres.nhs.uk/applications/guidance/governance-and-directives/?entryid62=131341 – Governance Arrangements for Research Ethics Committees
- http://www.nres.nhs.uk/EasySiteWeb/GatewayLink.aspx?alId=134016 - NRES algorithm "Does my project require review by a Research Ethics Committee"?
- http://www.hta.gov.uk/policiesandcodesofpractice/codesofpractice.cfm - Human Tissue Authority Code of Practice
- http://www.hta.gov.uk – Human Tissue Authority website
- http://www.rsclearn.mrc.ac.uk/MRC_HumanTissueAct/player.html - Medical Research Council online training course for Human Tissue Act.

What to do next:

- **If you have answered NO to questions 1-18 (inclusive) and YES to questions 19-25 (inclusive)**, you do not need to complete the Full Research Ethics Approval Form (Part B). Please keep this form for your records, and do not submit to **Faculty Research Ethics Committee** (FREC) unless you require ethical consideration of your study, regardless of ethical implications, by an external body (question 26 has been answered YES). **A list of Faculty contacts is below.**
- **If you have answered YES to any of the questions 1-18 (inclusive) or NO to any of the questions 19-25** the Full Research Ethics Approval Form (Part B) MUST be submitted including Cover Sheet, Part A and Part B of the application form plus any required supplementary documents to the Secretary of the relevant Faculty Research Ethics Committee (FREC). **A list of Faculty contacts is below.**
- If you are applying for external Ethical Approval, please send a *copy* of the Conditions/Approvals letters to the **University Research Ethics Committee** (UREC) Secretary (this may include the original ethical application(s)). Where the external ethics committee/body has equal standing or primary jurisdiction, e.g. another University Research Ethics Committee or a National Research Ethics Committee, any approval will normally be received and noted by the University of Westminster Research Ethics Committee and further consideration may not be required. Where the external committee does not have equal or higher standing than the University Committee then the full ethical approval process at the university may still be required. Additional institutional compliance issues may need consideration by UREC.
- All Applications (dated, signed and authorised) and supplementary information or External Approvals should be sent to the University Research Ethics Committee (UREC) Secretary in *electronic format with a version number, document name and date and the Principal Investigator (or Undergraduate/Postgraduate Taught Student) name. On receipt your application will be issued a unique reference number*
- All new Applications should be submitted to a Research Ethics Committee (FREC or UREC) Secretary a minimum of 10 working days in advance of the Committee meeting date (earlier submission is recommended so that applications can be pre-vetted and obvious issues addressed before the application is considered by the Committee).

Contact details:

University Research Ethics Committee	**Professor Graham Megson**	**Huzma Kelly**

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